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The Editors

Applied Spatial Analysis and Policy

Dear Editors,

We are pleased to submit the enclosed manuscript, “**Multidimensional Racial Residential Segregation across Brazilian Cities: A Nationwide Assessment with 2022 Census Data,**” for consideration as a research article in *Applied Spatial Analysis and Policy*.

Context and importance. Residential segregation by race is one of the most durable axes of urban inequality in Brazil, conditioning access to services, schooling, environmental quality and health. Yet almost all quantitative work on the subject — nationwide and metropolitan alike — rests on a single measure, the index of dissimilarity, and therefore on just one of the five dimensions of segregation distinguished by Massey and Denton; spatially explicit measures are rarely used, and most studies still draw on the 2010 Census or earlier. This paper provides the first description of racial residential segregation in Brazil that is at once **nationwide** (all 319 municipalities above 100,000 inhabitants), **multidimensional** (nine indices spanning all five dimensions, five of them spatially explicit), and based on the **2022 Census**, the most recent tract-level data available. The analysis runs through a fully open and reproducible pipeline.

The results show that the choice of measure is consequential. The four evenness indices are near-redundant, and on that dimension segregation is concentrated in the South and Southeast — confirming and extending the reference 2010 study. But the other dimensions do not follow: the exposure dimension inverts the regional ranking, being highest in the North and Northeast for compositional reasons, and the identity of “the most segregated city” changes entirely with the dimension chosen (the dissimilarity and isolation rankings correlate at a Kendall’s τ of about -0.26). The Brazilian urban system varies along at least two, and arguably three, independent axes. A one-index characterisation is therefore incomplete, and — because the dimensions implicate different policy levers — the measurement choice has practical stakes for how cities monitor and address segregation.

Fit with the journal. The paper is an application of spatial analysis at national scale with an explicit policy dimension, which is the core of this journal’s scope. It does not propose a new method; it uses established spatially explicit indices to produce a spatially resolved, policy-relevant description of an entire urban system, and it argues concretely for a multidimensional monitoring dashboard and for region-specific policy responses (service equalisation in the majority-minority North and Northeast; land-use and housing instruments in the consolidated metropolitan cores of the South and Southeast). Closely related work on the sensitivity of spatial segregation indices and on comparative spatial segregation analytics has appeared in this journal and in adjacent spatial-analysis venues, and the readership that uses these indices is the audience this paper is written for. The reproducible pipeline — package, driver scripts, pinned environment, and one-command regeneration of every figure and table — reflects the journal’s emphasis on transparent, applied spatial analysis.

Declarations. The manuscript is original, has not been published previously, and is not under consideration by any other journal. The uploaded manuscript is anonymised for double-blind review. The study uses only publicly available data aggregated by census tract, with no human participants, so ethics approval is not applicable. The authors have no competing interests to declare, and no funding was received for this work. Statistical inference is deliberately outside the scope of the paper and is identified as future work; the contribution is a point-estimation description.

Thank you for considering this submission. We look forward to the reviewers' comments.

Yours sincerely,

The Authors